



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice

21 September 2011

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **Answer Sheet**.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the Question Paper.

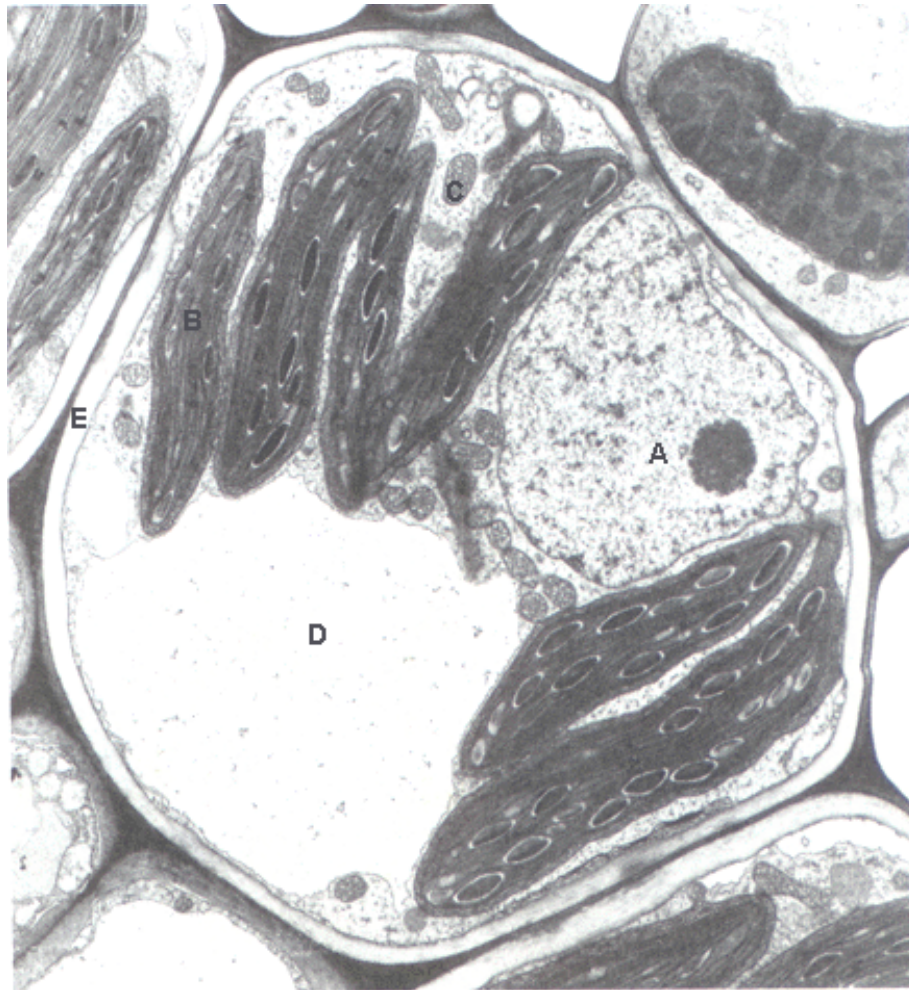
Calculators may be used.

This Question Paper consists of **22** printed pages.

For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

1. **Fig. 1** represents part of a plant cell.



Source: <http://library.thinkquest.org/>

Fig. 1

If the cell has been growing in an environment in which its sole source of nitrogen had been labelled radioactively, we would expect to find radioactivity in structure(s)

- A** **A** only.
- B** **A, B** and **C** only.
- C** **B, C** and **D** only.
- D** **C, D** and **E** only.

2. **Fig. 2** is a diagram of the secretory pathway from the endoplasmic reticulum (ER), through the Golgi apparatus, to the plasma membrane.

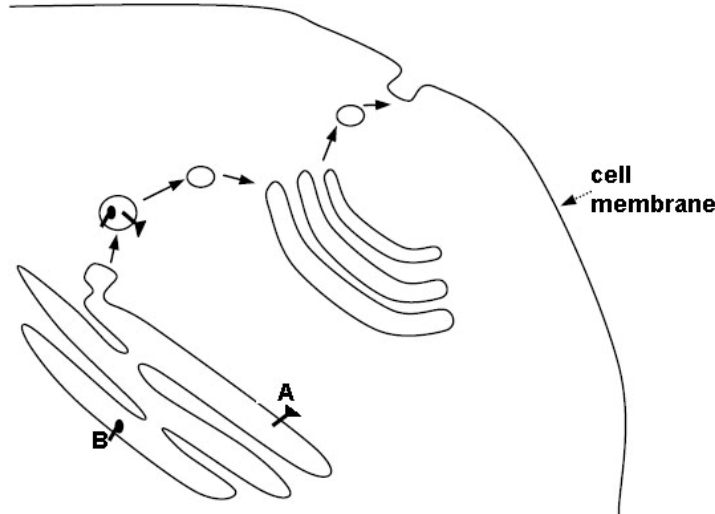
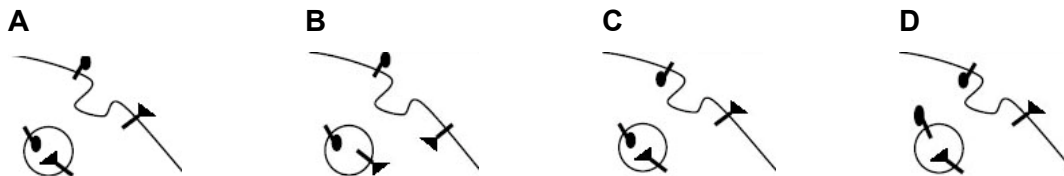


Fig. 2

Which of the following diagrams correctly represents the orientations of proteins **A** and **B** upon reaching the secretory vesicle and the plasma membrane?



3. The four acids shown in **Fig. 3** form part of an enzyme-catalysed sequence.

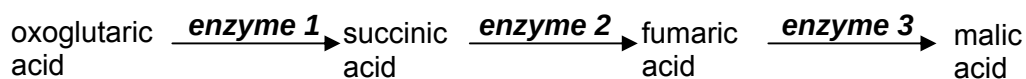


Fig. 3

The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid.

Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

- A** It is an inhibitor of **enzyme 1**.
- B** It catalyses the formation of succinic acid.
- C** It is an inhibitor of **enzyme 2**.
- D** It reacts with fumaric acid.

4. **Fig. 4.1** is an electronmicrograph showing a cell at a particular stage during the cell cycle.

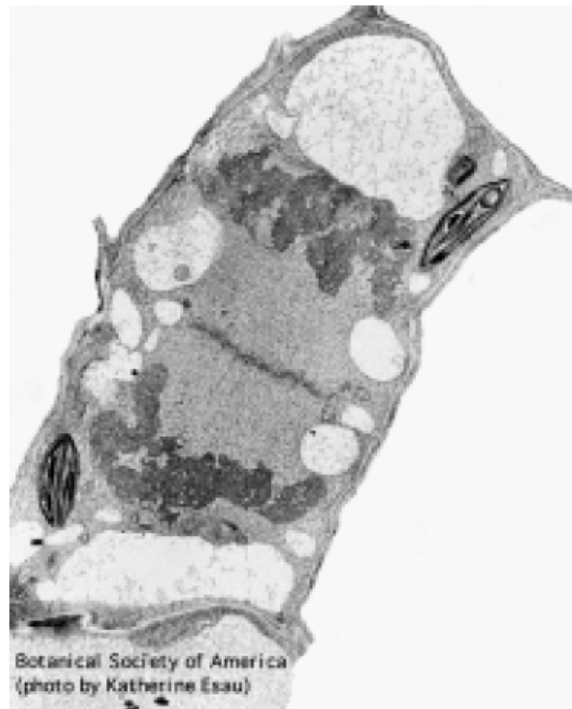


Fig. 4.1

Fig. 4.2 shows the mass of DNA within a cell during the cell cycle.

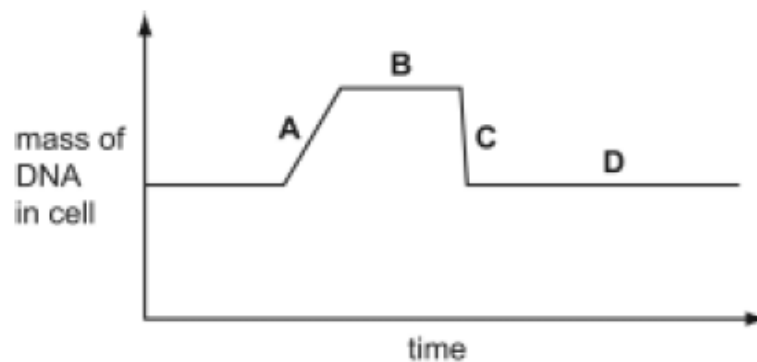


Fig. 4.2

When does the stage shown in **Fig. 4.1** occur?

5. A cell in the G1 phase has two homologous pairs of chromosomes. It undergoes a mitotic division, followed by meiosis. At the end of meiosis II, what is the sum of all the products of these divisions in all cells?
- A 16 chromosomes, 16 chromatids, 8 cells, 4 times as many alleles
 - B 16 chromosomes, 16 chromatids, 4 cells, 4 times as many alleles
 - C 8 chromosomes, 16 chromatids, 8 cells, 2 times as many alleles
 - D 8 chromosomes, 16 chromatids, 16 cells, 4 times as many alleles
6. Fluorescence *in situ* hybridisation (FISH) can be used to detect differences in rRNA sequences among individuals in a bacterial population. To search for particular rRNA sequences, which of the following molecular probes would be appropriate?
- A a mRNA probe complementary to the rRNA sequence
 - B a rRNA probe complementary to the sequence being searched for
 - C a single-stranded DNA probe containing the rRNA genes that can bind with the rRNA sequences
 - D a double-stranded DNA probe containing the rRNA genes that code for the rRNA sequences being searched for
7. As shown in **Fig. 7**, there are two bidirectional replication forks per replication bubble normally. If a mutant exists such that it has only one functional fork per replication bubble, which of the following would happen?

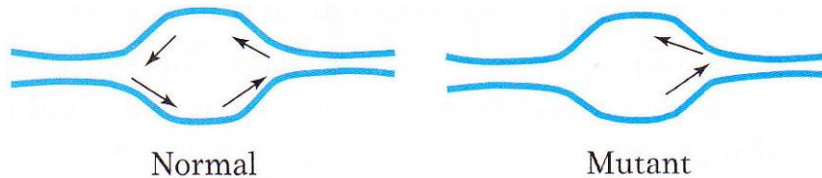


Fig. 7

- A DNA synthesis might take longer, but otherwise there would be no noticeable difference.
- B The DNA fragments from the two daughter strands could become mixed, producing possible mutations.
- C DNA synthesis would not take place because the appropriate enzymes to carry out discontinuous replication for both daughter strands would not be present.
- D DNA synthesis would not take place because the entire length of the chromosome would have to be unwound before both strands could be replicated in a discontinuous fashion.

8. Which of the following correctly matches the types of RNA molecules with their descriptions?

carries the information for the synthesis of one or more proteins

binds to an amino acid and guides it to the correct position on the ribosome

- | | | |
|----------|---------------|---------------|
| A | transfer RNA | messenger RNA |
| B | ribosomal RNA | messenger RNA |
| C | messenger RNA | transfer RNA |
| D | messenger RNA | ribosomal RNA |

9. Fig. 9 shows the results of two types of gene mutation.

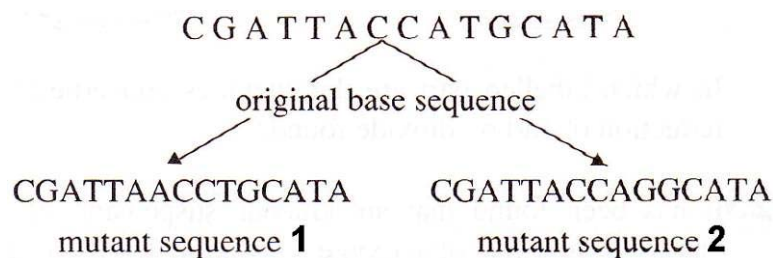


Fig. 9

Which types of gene mutation produce mutant sequences 1 and 2?

mutant sequence 1

mutant sequence 2

- | | | |
|----------|--------------|--------------|
| A | deletion | insertion |
| B | insertion | deletion |
| C | inversion | substitution |
| D | substitution | insertion |

10. Which statements correctly describe the transfer of DNA from one bacterium to another?

- 1** Both chromosomal and non-chromosomal DNA are transferred to daughter cells via binary fission.
- 2** Bacteriophage transfers DNA from one bacterium to another during specialised transduction.
- 3** Single strand of F plasmid transferred from one bacterium to another during conjugation.
- 4** Bacterial cell takes up foreign DNA from culture medium during generalised transduction.

- | | | | | | | | |
|----------|----------------|----------|----------------|----------|-------------------|----------|-------------------|
| A | 1 and 2 | B | 2 and 3 | C | 1, 2 and 3 | D | 1, 3 and 4 |
|----------|----------------|----------|----------------|----------|-------------------|----------|-------------------|

11. Tamiflu is an antiviral drug used to treat influenza A (H1N1) patients. It works by preventing progeny virions from detaching from infected cells.

Which of the following components of the influenza virus does Tamiflu act on?

- A Hemagglutinin
- B Matrix protein
- C Neuraminidase
- D Nucleoproteins

12. What happens when bacteriophage are grown in radioactive phosphorous?

- A Their DNA becomes radioactive.
- B Their proteins become radioactive.
- C They are no longer able to transform bacterial cells.
- D They transfer their radioactivity to bacterial chromosomes during infection.

13. The gene activity of the *lac* operon in four different *E. coli* strains was measured and the results are shown in **Table 13**.

Table 13

	<i>E. coli</i> strain	Genotype	Presence (✓) or absence (✗) of β-galactosidase activity	
			Lactose present	Lactose absent
1	E1	I ⁺ O ⁺ Z ⁺	✓	✗
2	E2	I ⁻ O ⁺ Z ⁺	✓	✓
3	E3	I ⁺ O ⁻ Z ⁺	✓	✓
4	E4	I ⁺ O ⁻ Z ⁻	✓	✗

Key: I⁺ – wild type (i.e. normal) allele for *lacI* gene
 I⁻ – allele for *lacI* gene which is not expressed
 Z⁺ – wild type (i.e. normal) allele for β-galactosidase gene of *lac* operon
 Z⁻ – allele for β-galactosidase gene which is not expressed
 O⁺ – functional operator of *lac* operon
 O⁻ – non-functional operator of *lac* operon

Which of the observation is **incorrect**?

- A **1**
- B **2**
- C **3**
- D **4**

14. The extent of condensation of chromatin plays an important role in the regulation of RNA transcription in eukaryotic cells. Chromatin-remodelling complexes are among proteins that control the physical state and functional activity of chromatin. In a study designed to determine the role of chromatin remodelling complexes, nucleosome core particles were prepared. Which of the following procedures was used to prepare the nucleosome core particles?
- A treatment of chromatin with absolute alcohol → centrifugation
 - B treatment of chromatin with SDS → centrifugation
 - C endonuclease digestion of chromatin → sucrose gradient centrifugation
 - D exonuclease digestion of chromatin → sucrose gradient centrifugation

Refer to **Fig. 15** below for **Questions 15** to **16**.

Transcription factor L (TFL) is examined for its DNA binding ability and its expression in different cell lines. The specific DNA binding sequence has been determined and is used as a probe in an electromobility (EMS) assay. Normally, short DNA sequences will run from one end of a gel to the other. When used as probes, these DNA sequences can bind to proteins that slow them down and prevent them from travelling through the gel.

Specific antibodies are also used in Western blot analysis. In this technique, antibodies bind to proteins that are run out on a gel to indicate the presence of a particular protein at a specific cellular location. The EMS and Western blot results for **TFL** activity in muscle cells and nerve cells are shown in **Fig. 15** below.

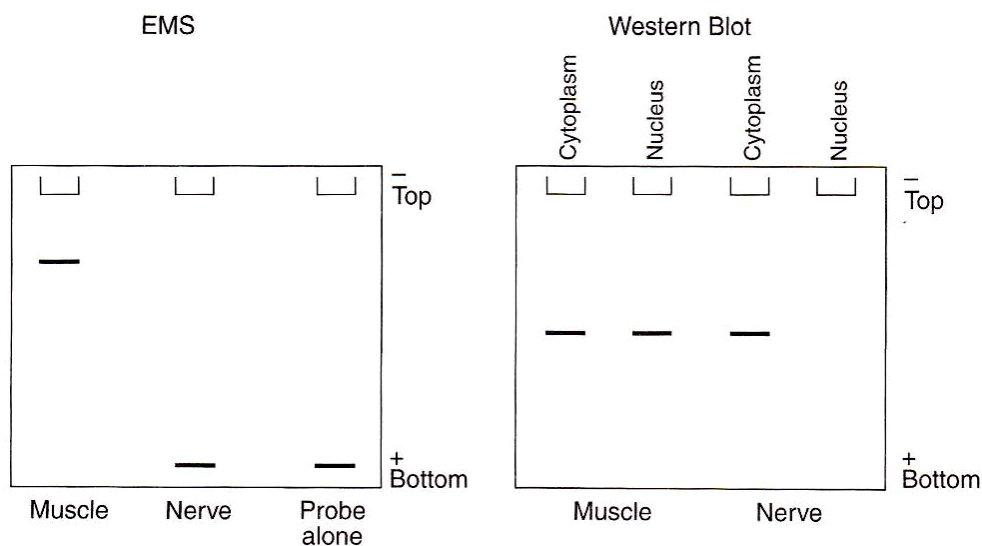
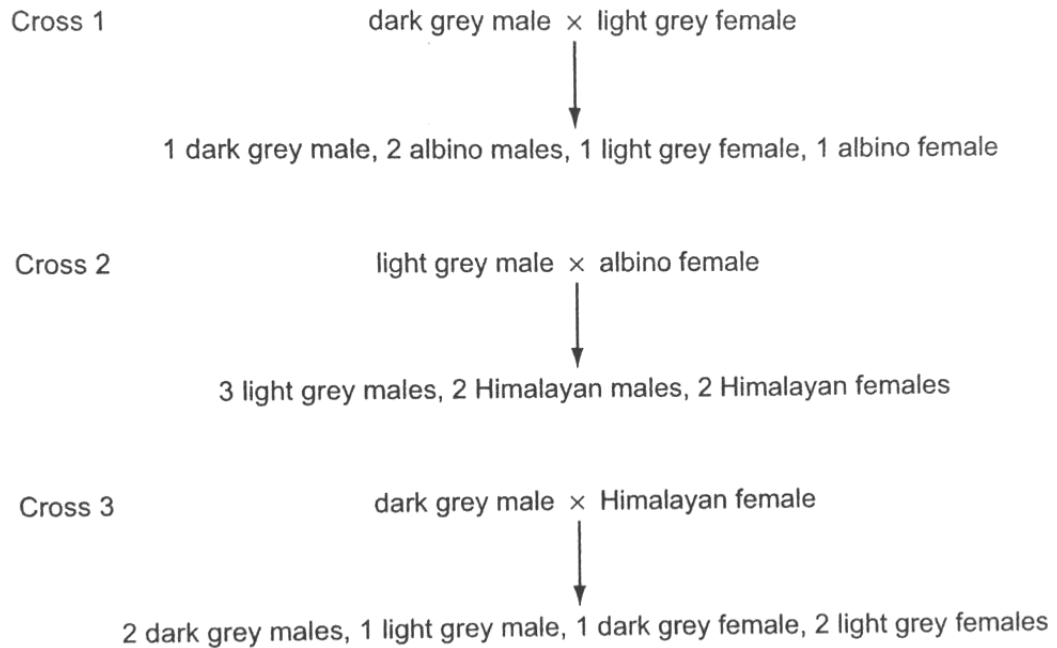


Fig. 15

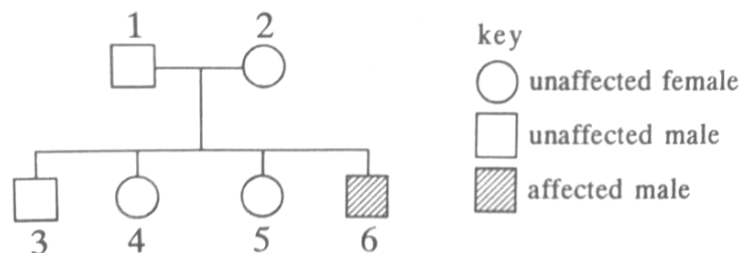
15. The EMS results suggest that
- A TFL is expressed only in nerve cells.
 - B TFL is expressed only in muscle cells.
 - C TFL binds DNA in muscle cells, but not in nerve cells.
 - D TFL binds DNA in nerve cells, but not in muscle cells.
16. The data from the experiment seem to suggest that the most likely mechanism for the regulation of TFL is
- A transcription.
 - B phosphorylation.
 - C translocation to the nucleus.
 - D translocation to the cytoplasm.
17. Which of the following best describes the process of alternative splicing?
- A various ways of reading nucleotides in groups of three from a start codon
 - B the succession of codons determined by reading nucleotides in groups of three from a specific start codon
 - C the reaction that removes introns and joins together exons in eukaryotic nuclear primary RNA transcripts
 - D various ways of splicing out introns in eukaryotic pre-mRNAs resulting in one gene producing several different mRNAs and protein products
18. Over 50 % of human tumours have developed an inactivating mutation in p53 activity. The lack of this activity contributes to tumour cell growth via which of the following mechanisms?
- A activation of MAP kinases
 - B increase in apoptotic events
 - C increase in DNA mutation rates
 - D increase in transcription-coupled DNA repair

19. The diagram shows the results of a series of crosses in rabbits.



What explains the inheritance of the range of phenotypes shown by these crosses?

- A a pair of co-dominant alleles
- B one gene with multiple alleles
- C sex linkage of the allele for grey coat colour
- D two genes, each with a dominant and recessive allele
20. The diagram shows the inheritance of haemophilia in a family.



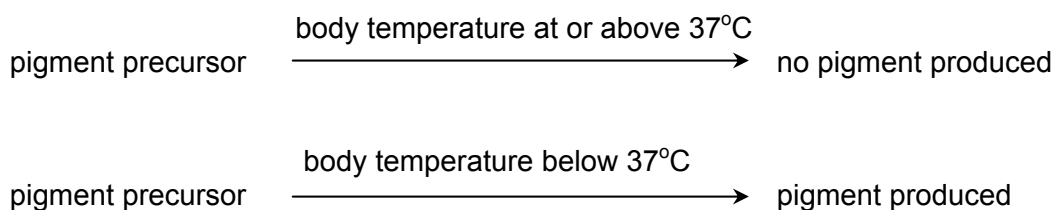
If daughter 4 married a normal male, what is the probability that their first child would suffer from haemophilia?

- A 0
- B 1 in 8
- C 1 in 4
- D 1 in 2

21. The following image is of a Siamese cat.



The colour produced in the fur of a Siamese cat is under the control of a tyrosinase enzyme that is sensitive to particular temperatures. A summary of the enzyme action is



The core body temperature of a cat is around 38°C .

It is likely that in London,

- A Siamese kittens have dark fur over their bodies when they are born.
- B Siamese cats that live outside a house show an increase in pigment during summer.
- C Siamese cats that live inside a house in winter have darker fur than those that live outside.
- D Siamese cats that live outside in winter have darker fur than Siamese cats in tropical regions.

22. Fig. 22 shows a human karyogram.



Fig. 22

What can be concluded from the karyotype provided?

- A** There was non-disjunction during meiosis I in the mother.
- B** There was non-disjunction during meiosis II in the father.
- C** One contributory gamete to the zygote is an egg with an X and a Y chromosome.
- D** One contributory gamete to the zygote is a sperm containing two X chromosomes.

23. Test crossing some *Drosophila* gave the following results.

Red eye, normal wing	58
Purple eye, vestigial wing	42
Red eye, vestigial wing	38
Purple eye, normal wing	62

The ratio of phenotypes expected for such a dihybrid test cross is 1:1:1:1.

A χ^2 (chi-squared) test was performed to test the significance of the difference between the observed and expected results.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

Where Σ = 'sum of...'

ν = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

degrees of freedom	probability, p						
	0.50	0.20	0.10	0.05	0.02	0.01	0.001
1	0.46	1.64	2.71	3.84	5.41	6.64	10.38
2	1.39	3.22	4.61	5.99	7.82	9.21	13.82
3	2.37	4.64	6.25	7.82	9.84	11.35	16.27
4	3.36	5.59	7.78	9.49	11.67	13.28	18.47

Which combination correctly describes the result of the χ^2 test?

	probability	significance of difference between observed and expected results	loci of genes
A	< 0.05	not significant	different chromosomes
B	< 0.05	significant	different chromosomes
C	> 0.05	not significant	same chromosome
D	> 0.05	significant	same chromosome

24. A student, **X**, looked after a plant. Another student, **Y**, looked after another plant of the same species. Each student followed the same instructions to set up apparatus to take cuttings from their plant and grow the cuttings next to the plant from which the cutting had been taken.

Fig. 24 shows the results after one week.

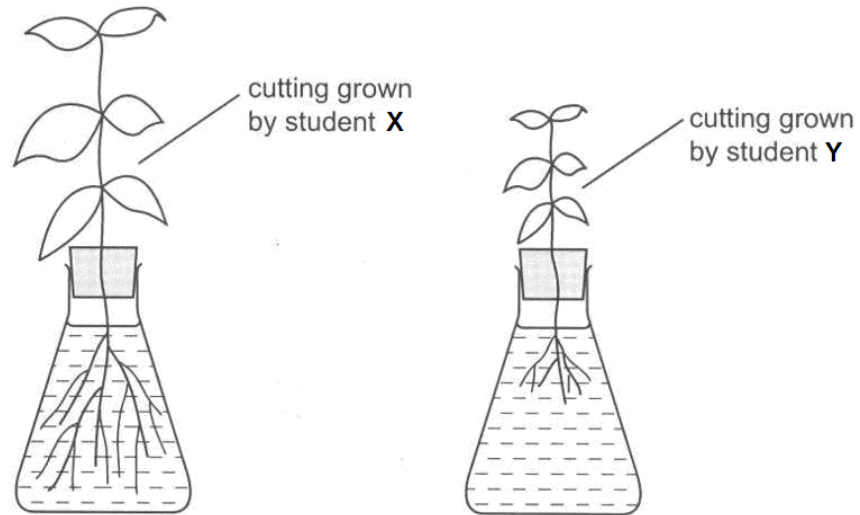


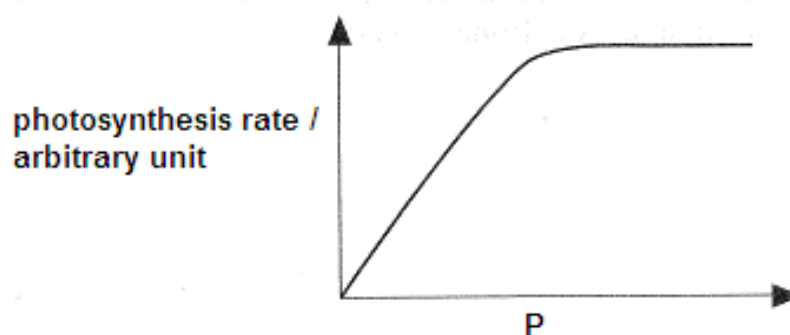
Fig. 24

Which factors could have caused the different appearance of the two cuttings?

- 1 phenotypic variation due to environment
- 2 genetically different cuttings
- 3 mutation due to environment
- 4 genotypic variation due to environment

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 1, 3 and 4

25. The following graph shows the relationship between the rates of photosynthesis with environmental factor **P**.



P most probably represents

- 1 carbon dioxide concentration.
- 2 oxygen concentration.
- 3 light intensity.
- 4 temperature.

A 1 and 2 **B** 1 and 3 **C** 1 and 4 **D** 1, 3 and 4

26. It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which colours of light are required for maximum oxygen evolution?

	Type of compound	Colours of light at which maximum evolution occurs
A	Electron acceptor	Blue and green
B	Electron acceptor	Blue and red
C	Electron donor	Blue and green
D	Electron donor	Blue and red

27. The reaction below occurs in all aerobic cells.

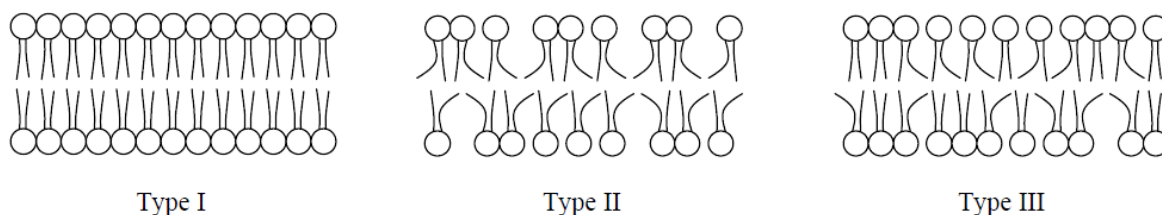


Which of the following statements are **true** of the above reaction?

- 1 The reaction happens in the cytoplasm of the cells.
- 2 The above reaction involved oxidative decarboxylation.
- 3 2H are released and received by NAD^+ .
- 4 ATP energy is used in the above reaction.
- 5 Carbon dioxide is released during this reaction.

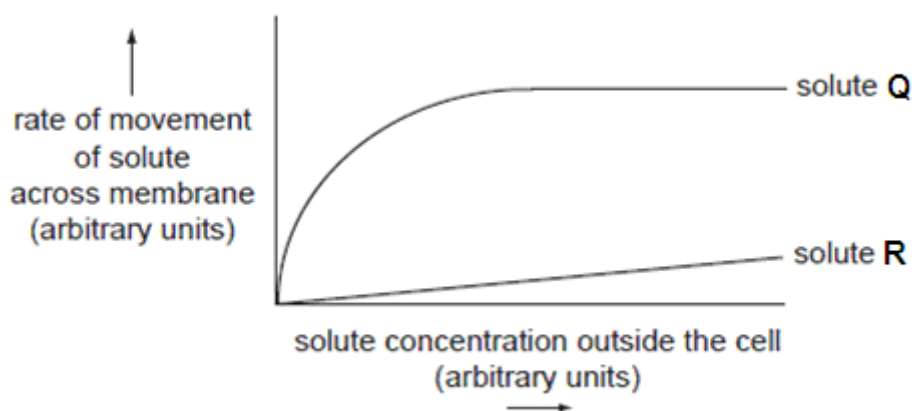
A 1 and 3 **B** 2 and 4 **C** 1, 2 and 4 **D** 2, 3 and 5

28. A mutated strain of the bacterium *Escherichia coli* was found to be incapable of incorporating unsaturated phospholipids into its plasma membrane. Which of the following correctly depicts and describes the membrane of such a bacteria?



- A** The membrane would appear as Type I and would be more fluid at low temperatures.
- B** The membrane would appear as Type I and would be less fluid at low temperatures.
- C** The membrane would appear as Type II and would be more fluid at low temperatures.
- D** The membrane would appear as Type III and would be no different from normal bacterial membranes.
29. Solute **Q** moves across the plasma membrane via protein channels. Solute **R** moves across the plasma membrane by simple diffusion. The rate of movement of each solute into a cell is recorded and graphed.

The results are shown in the following graph.



It would be reasonable to conclude that

- A** solute **Q** is lipid soluble.
- B** solute **Q** saturates the protein channels.
- C** the gradient of the graph for solute **R** will increase as the temperature decreases.
- D** as solute **R** is metabolised within a cell, its rate of movement across the membrane decreases.

30. One action of the hormone glucagon is to cause the activation of adenylyl cyclase in liver cells. Adenylyl cyclase is an enzyme that converts ATP into the messenger molecule cyclic AMP (cAMP), the presence of which triggers a cascade of events within the cell that leads to the breakdown of glycogen and the subsequent release of glucose into the blood stream. This pathway is summarised in **Fig. 30**.

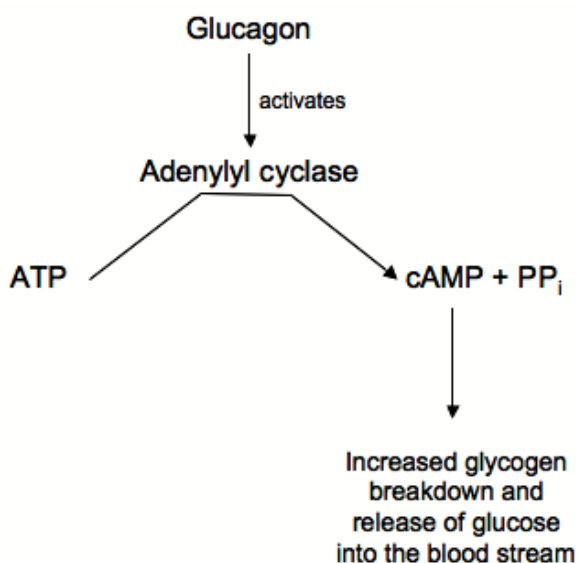


Fig. 30

When glucagon is removed adenylyl cyclase becomes inactive again and ceases to convert ATP to cAMP. The enzyme phosphodiesterase converts cAMP into AMP. The following table summarises an experiment measuring glucose release by liver cell cultures under various conditions.

culture	condition(s)	observations
1	no addition	minimal glucose release
2	glucagon added and later removed	glucose released into media, rapid reduction in release after glucagon removed
3	caffeine added	minimal glucose release
4	glucagon and caffeine added, glucagon then removed but caffeine maintained	glucose released into media, slow reduction in release after glucagon removed.

Based on these observations, which of the following is most likely?

- A Caffeine stimulates synthesis of cAMP.
- B Caffeine prevents glucagon breakdown.
- C Caffeine binds the active site of adenylyl cyclase.
- D Caffeine inhibits the action of phosphodiesterase.

31. Tetrodotoxin, an extremely potent poison produced by the puffer fish, binds tightly to voltage-gated sodium channels and blocks the flow of sodium ions but does not affect either potassium or chloride ion channels. Which phase of an action potential does Tetrodotoxin directly block?

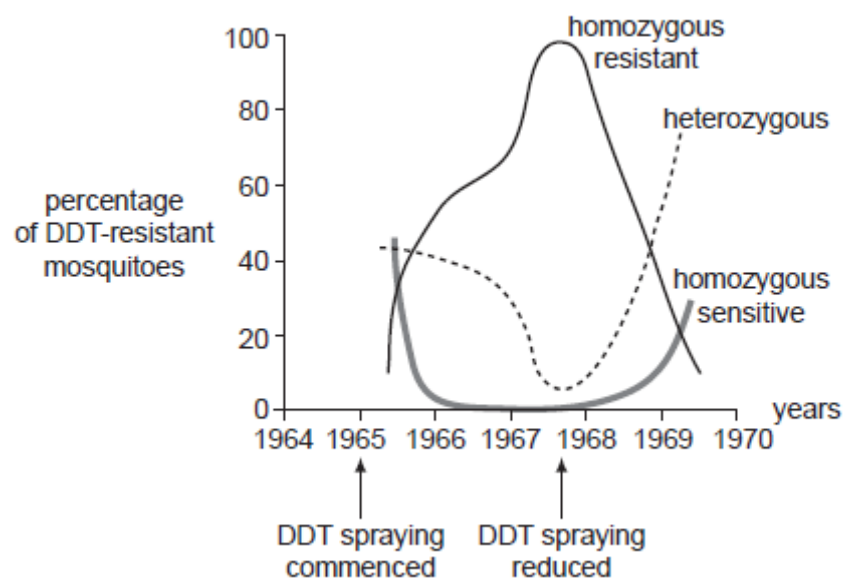
A depolarisation
B repolarisation
C hyperpolarisation
D refractory period

32. In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles.

allele 1 : resistant to DDT

allele 2 : sensitive to DDT

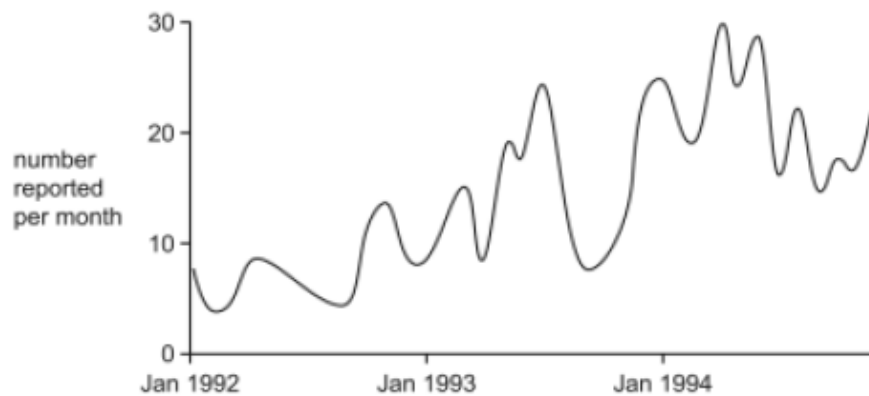
Over several years genotypic frequencies were measured in a population of mosquito larvae. The graph below shows the results.



Analysis of the graph reveals that in the population

- A when spraying levels declined, heterozygous advantage occurred.
B there were no alleles for sensitivity present in the population in 1967.
C the number of alleles for resistance was equal to the number for sensitivity in 1966.
D the homozygous resistant genotype was unable to produce offspring at low spraying levels.

33. The graph shows the change in number of bacterial samples from some New York hospitals that were resistant to the antibiotic vancomycin in 1992–4. Forty samples were taken each month from randomly selected patients who had become infected with bacteria in hospital.



Which of the following most accurately describes the cause of the changes in the frequency of the vancomycin resistant phenotype that occurred?

- A effect of artificial selection
- B effect of natural selection
- C purely due to genetic drift
- D purely due to random sampling effects

34. The ground finch, *Geospiza fortis*, is a species of bird which lives on a small isolated island.

These finches feed on seeds of different sizes from different species of plants. The finches show variation in the size of their beaks. Birds with larger beaks can eat large and small seeds. Birds with smaller beaks are only able to eat small seeds.

In 1977 there was a severe drought on the island. This killed many species of plants that the finches fed on. One species of food plant did survive and this produced large seeds.

The graphs show the distribution of beak sizes of the finch population before and after the drought. Beak size was measured by the depth of the beak, as shown in **Fig. 34**.

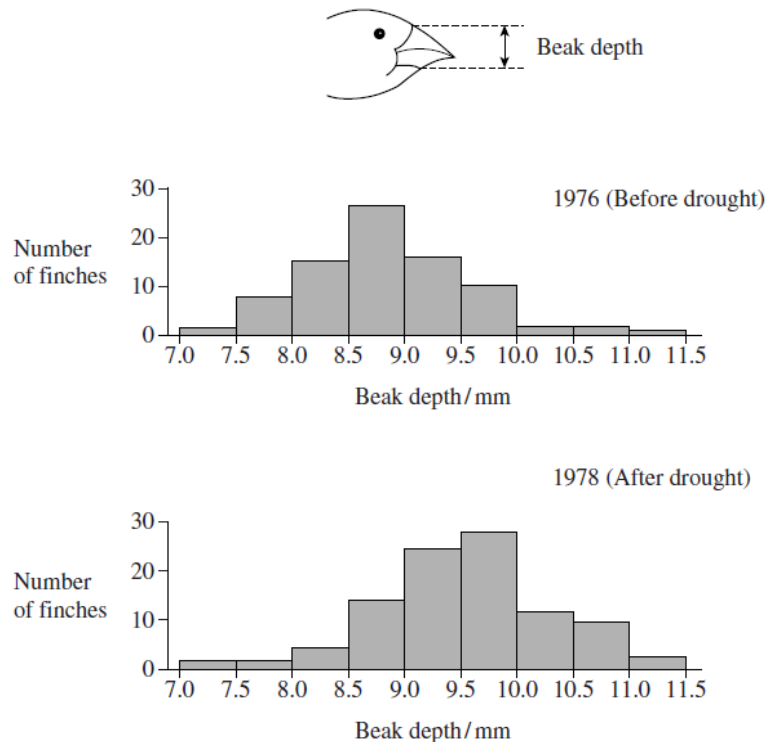


Fig. 34

Which type of selection and inheritance does this show?

	<i>type of selection</i>	<i>inheritance</i>
A	directional	polygenic
B	directional	multiple alleles
C	stabilizing	polygenic
D	stabilising	multiple alleles

35. Which primers should be paired to amplify, by PCR, the DNA flanked by the two indicated sequences?

5' – GGAATTCGT -- // -- AATGCTACC – 3'

- A ACGAATTCC & AATCGTACC B AATGCTACC & GGAATTCGT
C GGTAGCATT & GGAATTCGT D ACGAATTCC & GGTAGCAAT
36. A family exhibits a rare autosomal recessive disease, which is caused by a point mutation that abolishes an *Eco*R1 restriction site in the genome. To ascertain the genotype of family members, researchers (i) amplify a 1.8 kb region of this area using PCR; (ii) cut the PCR products with *Eco*R1; and then (iii) separate the DNA fragments on an agarose gel. The results of such an experiment are shown in **Fig. 36** below, using DNA obtained from each of the five family members.

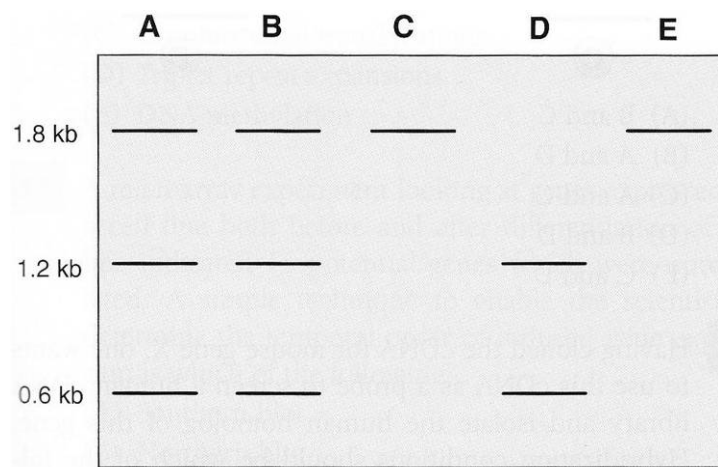


Fig. 36

Which family members have the disease?

- A A and B B C and E
C D only D A, B and D
37. Which of the following best explains why a genome project is not finished when the sequence has been completed?
- A Genomes change too fast and must constantly be updated.
B Without knowing the number, the function, and the location of genes within a genome, the sequence is not very useful.
C Individuals within a species are so unique that having a single sequence is insufficient to characterise a species' genome.
D Many sections of a genome are too difficult to sequence and have not actually been included in 'complete' genomes.

- P** inserting genes for vitamin A production into rice
- Q** inserting genes for pest resistance into cotton
- R** inserting genes for herbicide resistance into oilseed rape
- S** inserting genes for nitrogen fixation into soya beans

B **Q** and **R**

D Q and P

- Which of the following properties of stem cells makes them particularly useful in this research?

- D** 2, 3 and 4

- Which of the following could not be the explanation behind the failure in the treatment?

- D** None of the above